

3(a)	<i>max 2 of:</i> 1 can, kill / control, weeds ; R kill, insects / pests 2 reduce competition / increase yield (of rape) ; 3 AVP ; e.g. manual weeding / hoeing, difficult / expensive <i>ref. to</i> glufosinate converted to non-toxic compound	2
3(b)(i)	circle of / circular, DNA ; I loop R single-stranded small / supplementary ;	2
3(b)(ii)	<i>max 3 of:</i> 1 small so can be taken up by, cells / bacteria ; 2 replicate, independently / fast ; A have <i>ori</i> / origin of replication / high copy number 3 (DNA) has restriction site(s) / can be cut by restriction enzymes ; A have polylinker 4 have, marker genes / genes for resistance (for screening) ; 5 AVP ; e.g. circular so, increased stability / reduced host cell degradation	3
3(b)(iii)	<i>max 2 of:</i> 1 <u>RNA polymerase</u> binds ; 2 so, transcription / mRNA synthesis, begins / occurs / allowed ; 3 AVP ; e.g. correct / template, strand is transcribed <i>ref. to</i> tissue-specific / inducible, expression	2
3(c)(i)	28 ;	1

Question	Answer	Marks
3(c)(ii)	<i>max 1 of:</i> spray with herbicide and, those that die did not have the <i>bar</i> gene / those that survive did have the <i>bar</i> gene ; add gene for fluorescence with <i>bar</i> gene and test plants under UV / use PCR with primer complementary to <i>bar</i> gene / use (gene) probe (on Southern blot) of electrophoresis gel ;	1
3(c)(iii)	<i>max 3 of:</i> <i>advantage of male sterile GM variety</i> 1 avoid transferring, <i>bar</i> / resistance, gene to wild, radish / relations ; ora 2 avoid superweeds ; ora 3 avoid <u>type 2</u> hybrids ; ora <i>disadvantage of type 2 hybrids (from GM variety that produces pollen)</i> 4 taller (than wild radish) ; A very tall / 88 cm / 95 cm 5 produce, more / many, seeds (than wild radish) ; A 3958 / 443 more 6 may (out)compete, wild radish / crops ;	3

Question	Answer	Marks
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